
Quiz 03

Duration: 60 minutes

Student Name:

Student ID:

Student Tutorial:

In solving the problems, we are going to use your GUC ID.

Given your GUC Number in the form AB-CDEFG,

EX. If your ID is 12-34567 then $A = 1$, $B = 2$, $C = 3$, $D = 4$, $E = 5$, $F = 6$, and $G = 7$

If your ID is 12-3456 then $A = 1$, $B = 2$, $C = 3$, $D = 4$, $E = 5$, $F = 6$, and $G = 0$

If your ID is 12-0456 then $A = 1$, $B = 2$, $C = 0$, $D = 4$, $E = 5$, $F = 6$, and $G = 0$

First, fill in your data below!

A =	B =	C =
D =	E =	F =
G =		

Quiz 03

Duration: 60 minutes

Quiz Regulations:

- The quiz starts Monday May 19, 2025 (4:00 pm), and the duration is 45 minutes for solving + 15 minutes for submission (form will be closed at 5:00 pm sharp).
- At first, open the CMS, download the quiz, solve it on paper, and then fill in the form.
- The quiz answers should be submitted to the link:
https://docs.google.com/forms/d/e/1FAIpQLSd9_SXa4UbQ3FJ8OUAUwaj0PhJU8c_YnYmeMV54HFkEb0uViA/viewform?usp=dialog
Note that no answers will be accepted on emails!
- Please note that by the end of the quiz, you will be asked to scan and upload your solution on paper for reference but the main graded questions are the ones in the form.
- The quiz form is organized into 4 sections; the first one we are in already to enter your data, the second one is to get some data from your ID which we will use to solve the quiz, the third section is the quiz questions, the last section is to upload your scanned solution and submit.
- Please make sure that you answered all questions before submission.
- Make sure to press the submit button before the deadline so that your answer counts.
- Late submissions will not be considered and will cause cancellation of the quiz.
- Each student is allowed to submit ONLY ONCE. Multiple submissions will cause the cancellation of the quiz.

Important Notes:

- It is important that you provide your answers in the correct number of bits, otherwise it will be marked as wrong answer.
- Note that when a Hexadecimal number is needed, you can ignore the (0x) while entering your answer
Example if the answer is 0x1234 write in the form only 1234.
- Only write down numbers in the form answer boxes no need to add any units, but you must respect the units stated in each question.
- Note that if you miss entered your ID digits correctly, this will result in a ZERO in the quiz.

Quiz 03

Duration: 60 minutes

Virtual Memory Design

- A virtual memory is designed with a default word size of 1 Byte.
- The VM is composed of a number of addressable **pages** where each page size is 4^{D+2} KB where D is a digit from your ID number filled in the table in page 1.
- The VM has logical address with number of bits m where m is equal to one of the following 5 cases depends on D, E, F and G digits from your ID:

$$m = \begin{cases} E + F + G + 30 & , & (D + 10) \% 5 = 0 \\ E + F + G + 30 + 1, & (D + 10) \% 5 = 1 \\ E + F + G + 30 + 2, & (D + 10) \% 5 = 2 \\ E + F + G + 30 + 3, & (D + 10) \% 5 = 3 \\ E + F + G + 30 + 4, & (D + 10) \% 5 = 4 \end{cases}$$

Quiz 03

Duration: 60 minutes

Answer the questions below

Question 1: General Questions

- Q1.1. What is the **virtual** address space size (**in GB**)?
- Q1.2. What is the page address offset bits number?
- Q1.3. What is the Virtual Page Number (VPN) bits number?
- Q1.4. What is the number of pages in the VM?
- Q1.5. What is the total VM size (**in GB**)?
- Q1.6. What is the number of words per page?

Question 2: For the below questions, provide all the requirements in **Hexadecimal format**

- Q2.1. For the word with **decimal** **logical** address ($E \times 30$) (**E from the ID digits**):
 - a. What is the page number?
 - b. What is the offset?
- Q2.2. For the word with **decimal** **logical** address ($\text{floor}\left(\frac{M-5}{2}\right)$) (**M is the virtual address space size**):
 - a. What is the page number?
 - b. What is the offset?

Question 3: For the below questions, provide all the requirements in **Hexadecimal format**

- Q3.1. For the word stored in **decimal** **virtual** address (2), what is its corresponding **physical** address? Mapped to **decimal** PFN = $(A + B)$ (**A and B from the ID digits**)
- Q3.2. For the word stored in **decimal** **virtual** address ($M - 5$), what is the corresponding **physical** address? Mapped to **decimal** PFN = $(C + E)$ (**C and E from the ID digits**)
- Q3.3. For the word stored in **decimal** **virtual** address ($\text{floor}\left(\frac{M-5}{2}\right)$), what is the corresponding **physical** address? Mapped to **decimal** PFN = $(A + D)$ (**A and D from the ID digits**)